Relational Time-of-Arrival as a Covariant Field Observable: From Event Currents to a Continuum Clock Limit

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Abstract

Time-of-arrival (TOA) in quantum theory sits at an awkward intersection: Pauli-type arguments obstruct a self-adjoint operator canonically conjugate to a semibounded Hamiltonian, while relativistic localization makes "the" time parameter detector- and frame-dependent. We propose a covariant alternative that treats TOA as a relational field observable rather than an operator. The construction uses two conserved ingredients: a matter current $\boldsymbol{J^\mu}$ that records detector crossings through a world-tube $\boldsymbol{\Sigma}$, and an event-count current $\boldsymbol{j^\mu_{\mathrm{ev}}}$ whose integral defines a discrete clock $\boldsymbol{N_{\mathrm{ev}}}$. Coarse-graining $\boldsymbol{j^\mu_{\mathrm{ev}}}$ over a scale $\boldsymbol{\ell}$ yields an infrared clock field $\boldsymbol{T(x)}$, and TOA becomes the flux through $\boldsymbol{\Sigma}$ conditioned on the local clock reading $\boldsymbol{T|_\Sigma}$. In a $\boldsymbol{1+1}$-dimensional example with a narrowband Klein–Gordon wavepacket and a gapped quadratic clock EFT, the TOA density reduces to a convolution of the semiclassical flux profile with the clock readout distribution. This predicts a calculable arrival-time broadening set by the clock variance, while massive clock correlations impose an exponentially suppressed envelope on contributions that would require spacelike clock correlations. The result is a TOA observable whose dependence on detector details is compressed into the geometry of $\boldsymbol{\Sigma}$, the effective flux $\boldsymbol{J^\mu}$, and the clock-sector correlators, together with a controlled UV-to-IR matching between event counts and continuum time.

Keywords: relational time, time of arrival, quantum time observables, covariant observables, event currents, quantum field theory

Notation summary

J^μ	conserved matter current (detector flux)
$j^\mu_{\rm ev}$	conserved event-count current (clock sector)
$N_{\rm ev}$	discrete event-count clock
$T(x)$ or $\chi(x)$	continuum clock field (IR limit)
Σ	detector world-tube
$\mathcal{F}$	detector flux functional $n_\mu J^\mu$
$(\cdot)_\ell$	coarse-graining over spacetime scale ℓ

1 Introduction

Time is usually supplied to quantum theory as an external parameter. Operationally, however, time is read from physical degrees of freedom, and in relativistic settings those degrees of freedom are local. This mismatch is visible already in nonrelativistic quantum mechanics: Pauli-type objections constrain the existence of a self-adjoint time operator canonically conjugate to a semibounded Hamiltonian in generic settings [1, 4]. In quantum field theory the situation is sharper, because localization, causality, and detector modelling enter from the start [4, 5].

Many successful TOA constructions therefore adopt explicitly operational definitions: POVM-based prescriptions such as Kijowski's [2] and later developments [3, 4], as well as detector models and measurement couplings [5]. These approaches produce useful arrival-time statistics, but typically tie the observable to a chosen clock variable, a preferred slicing, or detector microphysics.

In this paper we take a different starting point. We define TOA as a *relational field observable*: detector crossings are binned by a *physical clock field* constructed from a conserved event-count current. Concretely, we introduce (i) a conserved matter current J^μ that determines flux through a detector world-tube Σ, and (ii) an effective conserved current $j^\mu_{\rm ev}$ whose integral counts localized detector-relevant events and thereby defines a discrete clock. A controlled coarse-graining over a scale ℓ produces an infrared clock field $T(x)$, which plays the role of a continuum time variable on scales $\gg \ell$.

The payoff is twofold. First, the TOA observable is manifestly covariant and does not require postulating a self-adjoint time operator. Second, the clock sector carries dynamics: an effective field theory for T introduces correlation scales that enter arrival-time statistics in a parameter-controlled way.

Relation to existing time-of-arrival formalisms.

The TOA literature offers several workable definitions—POVM prescriptions, explicit detector models, and path-integral or histories-based constructions. Our goal is not to compete with their operational success, but to isolate a covariant object that can be stated directly at the field level. The key move is to treat the clock as part of the physics: TOA is defined by conditioning flux through Σ on a clock reading built from an event current. In that sense, "detector independence" is not achieved by abstracting away measurement, but by specifying the minimal conserved structures that any detector model must reduce to in the appropriate limit.

Relation to Page–Wootters and relational-clock frameworks.

The present construction is complementary to Page–Wootters-type and relational-clock frameworks rather than a replacement for them. In those approaches, time is typically recovered through conditional states or correlations between a clock subsystem and the rest of the system. Here, by contrast, the emphasis is not on Hilbert-space factorization or conditional amplitudes, but on a covariant field-level observable: the detector flux through a world-tube Σ conditioned on a local clock readout generated by an effective event current. The novelty of the present formulation is therefore structural rather than metaphysical: it packages relational time-of-arrival directly as a covariant detector functional with explicit UV-to-IR matching between discrete event counts and a continuum clock field, together with a calculable broadening formula in the clock-EFT regime. In this sense the construction is in dialogue with related perspectives on relational clocks, quantum reference frames, and operational notions of time [12–14], while keeping the main object of interest at the level of a covariant detector functional rather than a conditional state construction.

Rather than revising measurement postulates or postulating a generalized time operator, we keep the standard field-theoretic setting and change only what is taken to be primitive: TOA is defined as a relational field-functional built from conserved currents and a physical clock sector. The absence of a universal time operator is then not a gap to be patched, but a cue that TOA should be formulated relationally through correlations between physical degrees of freedom.

Scope and assumptions.

Throughout this work, the event current $j^\mu_{\rm ev}$ is treated as a *physical but model-agnostic* structure. It may be fundamental (e.g. a primitive detection or interaction current) or emergent (e.g. a coarse-grained description of localized spacetime processes such as interaction vertices, defect crossings, or excitation worldlines). The approximate conservation $\nabla_\mu j^\mu_{\rm ev} \approx 0$ is expected to hold whenever the relevant events are robust under smooth deformations on the scales of interest, with violations suppressed by the ultraviolet resolution scale.

The coarse-graining map $(\cdot)_\ell$ is assumed to be a local, causal smoothing operation over spacetime regions of characteristic size ℓ, preserving current conservation in the infrared. No specific kernel is fixed; admissible choices include covariant averaging, block-spin–type maps, or detector-induced temporal binning, provided they respect locality and current conservation at scales $\gg \ell$. Admissibility criteria for $j^\mu_{\rm ev}$ in a given experimental context are discussed further in Section 10.

2 Detector World-Tubes and Matter Flux

We work on a four-dimensional Lorentzian spacetime $(\mathcal{M}, g_{\mu\nu})$. A detector is modeled as a timelike world-tube $\Sigma \subset \mathcal{M}$, with intrinsic coordinates σ^a $(a = 1, 2, 3)$, induced metric h_{ab}, and future-directed unit normal n_μ.

Let J^μ denote a conserved matter current,

$$\nabla_\mu J^\mu = 0. \tag{2.1}$$

Definition 2.1 (Matter flux) The matter flux through Σ is

$$\mathcal{F}(\sigma) \equiv n_\mu J^\mu\big(X(\sigma)\big). \tag{2.2}$$

The total registered flux is

$$\mathcal{N} = \int_\Sigma \mathrm{d}^3\sigma\, \sqrt{|h|}\, \mathcal{F}(\sigma). \tag{2.3}$$

3 Event Currents and Discrete Clocks

Definition 3.1 (Effective event current) An *effective event current* is a vector field j^μ_{ev} satisfying

$$\nabla_\mu j^\mu_{\mathrm{ev}} \approx 0, \tag{3.1}$$

where equality may be exact or hold up to controlled corrections on the scales of interest.

Remark 3.1 (Status of the event current) The current j^μ_{ev} is not assumed to be unique or fundamental. Rather, it represents a coarse-grained, detector-relevant counting current associated with a specified measurement context. Different admissible choices of j^μ_{ev} correspond, in general, to different effective clock theories and therefore to different TOA statistics. This freedom is not hidden in the formalism: it is part of the model specification and is, in principle, experimentally distinguishable through the resulting clock correlators and arrival-time distributions.

Remark 3.2 (Concrete example) In a relativistic QFT scattering experiment, j^μ_{ev} may represent the effective flux of detector-registered interaction events, i.e. a spacetime current whose integral counts localized vertex/crossing events relevant to the measurement.

Definition 3.2 (Event count) The associated event count through a hypersurface Σ is

$$N_{\mathrm{ev}}(\Sigma) \equiv \int_\Sigma \mathrm{d}\Sigma_\mu\, j^\mu_{\mathrm{ev}}. \tag{3.2}$$

Proposition 3.1 *If $\nabla_\mu j^\mu_{\mathrm{ev}} = 0$, then $N_{\mathrm{ev}}(\Sigma)$ is invariant under smooth deformations of Σ that preserve its boundary.*

Sketch The difference between the integrals over two such hypersurfaces is the flux of $\nabla_\mu j^\mu_{\mathrm{ev}}$ through the interpolating spacetime region. The claim then follows from Stokes' theorem and current conservation. □

The discrete-event clock is

$$T_{\mathrm{disc}} \equiv N_{\mathrm{ev}}. \tag{3.3}$$

Conceptual hierarchy.

The logical structure of the construction may be summarized schematically as

$$\begin{array}{ccc} \text{event current } j^{\mu}_{\text{ev}} & \longrightarrow & \text{discrete clock } T_{\text{disc}} = N_{\text{ev}} \\ & & \downarrow \text{coarse-graining} \\ & & \text{continuum clock field } T(x) \end{array}$$

The event current is the fundamental object; discrete and continuous clocks are its ultraviolet and infrared descriptions, respectively.

4 Coarse-Graining and the Continuum Clock Limit

Let $(\cdot)_\ell$ denote coarse-graining over a spacetime scale ℓ. We assume that the coarse-grained event current is locally integrable in the sense that

$$\partial_{[\mu}\big(j^{\text{ev}}_{\nu]}\big)_\ell \approx 0 \tag{4.1}$$

on the infrared scales of interest. Under this condition, one may define a local scalar clock field $T(x)$ by

$$\partial_\mu T(x) \equiv \frac{1}{\nu_0}\,\big(j_{\text{ev},\mu}(x)\big)_\ell, \tag{4.2}$$

where ν_0 has dimensions of events per unit time.

Lemma 4.1 If $\nabla_\mu j^{\mu}_{\text{ev}} = 0$ and the local integrability condition (4.1) holds after coarse-graining, then $T(x)$ is locally well-defined up to an additive constant; global obstructions correspond to nontrivial topology.

Proposition 4.1 (IR equivalence) *At scales $\gg \ell$, event counts and clock differences match:*

$$N_{\text{ev}}(\Sigma) \approx \nu_0 \int_\Sigma \mathrm{d}\Sigma_\mu\, \partial^\mu T = \nu_0\, \Delta T, \tag{4.3}$$

where the last equality holds when Σ interpolates between hypersurfaces of constant T. This is the relevant situation when the detector region is used to compare two clock slices—for example, in the local detector setups considered below, where the clock field is sampled between successive level sets of the coarse-grained readout. For generic finite detectors in $3+1$ dimensions this exact identification need not hold globally, and the equality should then be understood as a local or idealized limit; extending the construction to fully finite detector geometries is part of the program outlined in the Conclusion.

5 Clock-Sector Effective Field Theory

For the purpose of a local EFT expansion in a Minkowski background, we choose an inertial coordinate patch and expand the clock field around a monotonically increasing background solution,

$$T(x) = T_0(x) + \tau(x), \qquad T_0(x) = t, \tag{5.1}$$

so that τ describes fluctuations around the reference clock profile.

A minimal stable quadratic EFT is

$$S_\tau = \frac{1}{2}\int \mathrm{d}^4x \left[(\partial_t\tau)^2 - c_\tau^2(\nabla\tau)^2 - \mu_\tau^2\tau^2\right]. \tag{5.2}$$

The parameter μ_τ sets a correlation scale in space and time.

In Appendix A, μ_τ is shown to control the exponential suppression envelope of spacelike clock correlations and the corresponding early-arrival bounds.

Remark 5.1 (Correlation scale) The mass parameter μ_τ sets the characteristic temporal and spatial correlation scale of the clock field. As shown explicitly in Appendix A, the same parameter controls the exponential suppression envelope of spacelike clock correlations and thereby motivates corresponding bounds on early-arrival contributions in TOA models where such contributions are clock-correlation controlled.

6 Relational Time-of-Arrival Observable

Definition 6.1 (Relational TOA distribution) The TOA probability density with respect to the clock field T is

$$p(\Theta) \equiv \frac{1}{\mathcal{N}}\int_\Sigma \mathrm{d}^3\sigma\,\sqrt{|h|}\,\mathcal{F}(\sigma)\,\delta\big(T_\Sigma(\sigma)-\Theta\big), \tag{6.1}$$

where $T_\Sigma(\sigma) \equiv T(X(\sigma))$.

Why this is nontrivial.

Equation (6.1) is not a relabeling of an external time coordinate. The clock $T(x)$ is a dynamical field (or effective field) with its own correlators, fixed by the clock-sector EFT. Those correlators enter TOA statistics through the averaging in Eq. (8.9) and lead to quantitative modifications—e.g. broadened arrival-time distributions and an μ_τ-controlled envelope for contributions that would require spacelike clock correlations (Appendix A).

This definition is covariant and compresses detector dependence into the geometry of Σ, the effective flux functional $\mathcal{F}$, and the clock-sector correlators.

7 Semiclassical Limit

In the semiclassical regime of sharply peaked wavepackets and weak clock fluctuations,

$$p(\Theta) \to \delta(\Theta - \Theta_{\mathrm{cl}}), \tag{7.1}$$

where Θ_{cl} matches the classical time-of-flight.

8 Worked Example in $1+1$ Dimensions

We now give an explicit $1+1$D computation (wavepacket + clock correlator). This section is intended as a concrete template; the same steps extend to $3+1$D with transverse integrals.

8.1 Setup: detector worldline and matter wavepacket

Work in Minkowski space with coordinates (t, x) and metric $\eta = \mathrm{diag}(1, -1)$. Model the detector as the timelike worldline at fixed $x = x_D$:

$$\Sigma : \quad X^\mu(\sigma) = (t = \sigma,\ x = x_D). \tag{8.1}$$

For a $1+1$D detector "world-tube", the induced measure reduces to $\mathrm{d}\sigma = \mathrm{d}t$.

Take a complex Klein–Gordon field ψ of mass m with conserved $U(1)$ current [6, 7]:

$$J^\mu = \frac{i}{2m}\left(\psi^* \partial^\mu \psi - \psi \partial^\mu \psi^*\right), \qquad \partial_\mu J^\mu = 0. \tag{8.2}$$

In $1+1$D, the spatial flux at the detector is $J^x(t, x_D)$. We take the incoming direction to be $+x$ and define the registered flux by

$$\mathcal{F}(t) \equiv J^x(t, x_D)\, \mathbb{1}_{\{J^x \geq 0\}}. \tag{8.3}$$

This positive-flux restriction is an operational assumption, standard in TOA constructions [2–4], and is appropriate in the narrowband, predominantly right-moving regime considered below. In that regime the negative-flux component is parametrically suppressed, so Eq. (8.3) modifies neither the normalization nor the convolution formula beyond subleading corrections. Outside that regime, backflow or bidirectional contributions must be treated explicitly and the simple positive-flux truncation need not be adequate.

Let the positive-frequency wavepacket be

$$\psi(t, x) = \int_{-\infty}^{+\infty} \frac{\mathrm{d}k}{\sqrt{2\pi}}\, a(k)\, e^{-i\omega_k t + ikx}, \qquad \omega_k = \sqrt{k^2 + m^2}. \tag{8.4}$$

Choose $a(k)$ narrowly peaked at $k_0 > 0$ with width $\Delta k \ll k_0$. Then the packet propagates with group velocity

$$v_g = \left.\frac{\partial \omega_k}{\partial k}\right|_{k_0} = \frac{k_0}{\omega_{k_0}} \in (0, 1). \tag{8.5}$$

8.2 Semiclassical approximation for the flux

Under the narrowband approximation, J^x reduces to the familiar form

$$J^x(t, x) \approx v_g\, |\psi(t, x)|^2, \tag{8.6}$$

up to corrections of order $\mathcal{O}((\Delta k/k_0)^2)$. A Gaussian $a(k)$ gives an approximately Gaussian arrival profile at $x = x_D$:

$$\mathcal{F}(t) \approx \mathcal{F}_0 \exp\left[-\frac{(t-t_{\rm cl})^2}{2\sigma_t^2}\right], \qquad t_{\rm cl} = \frac{x_D - x_0}{v_g}, \tag{8.7}$$

where x_0 is the initial packet center and $\sigma_t \sim 1/(v_g \Delta k)$. Normalize $\mathcal{N} = \int \mathrm{d}t\, \mathcal{F}(t)$.

8.3 Clock field on the detector and stationary correlator

Take the continuum clock field

$$T(t,x) = t + \tau(t,x). \tag{8.8}$$

The relational TOA density (6.1) becomes in $1+1$D

$$p(\Theta) = \frac{1}{\mathcal{N}} \int_{-\infty}^{+\infty} \mathrm{d}t\, \mathcal{F}(t) \left\langle \delta\big(t + \tau(t,x_D) - \Theta\big) \right\rangle_\tau, \tag{8.9}$$

where $\langle\cdot\rangle_\tau$ denotes averaging over clock fluctuations.

Assume the clock sector is in a stationary Gaussian state induced by the quadratic EFT (5.2). Then the single-point random variable $\tau_\ell(t,x_D)$ is Gaussian, with mean 0 and variance

$$\sigma^2_{\tau,\ell} \equiv \langle \tau_\ell(t,x_D)^2 \rangle = C_\ell(0,0), \tag{8.10}$$

where the two-point function is

$$C(\Delta t, \Delta x) \equiv \langle \tau(t,x)\tau(t+\Delta t, x+\Delta x)\rangle. \tag{8.11}$$

Here τ_ℓ denotes the clock field smeared over the detector's finite spacetime resolution scale ℓ. This coarse-grained variance is the physically relevant quantity entering the TOA distribution. In a continuum free-field description, the unsmeared coincidence limit $C(0,0)$ is UV-sensitive in $1+1$ dimensions, so the theory should be understood as defined either with an explicit ultraviolet cutoff Λ or, equivalently for present purposes, with a finite detector/coarse-graining scale ℓ. Accordingly, the observable variance is really $\sigma^2_{\tau,\ell} = \sigma^2_{\tau,\ell}(\mu_\tau)$, or equivalently $\sigma^2_\tau(\Lambda, \mu_\tau)$ in cutoff language.

In $1+1$D, for a gapped relativistic scalar ($c_\tau = 1$) at equal position $\Delta x = 0$, the stationary correlator along the detector worldline decays on the characteristic timescale μ_τ^{-1}. The timelike continuation of the massive correlator yields an oscillatory Bessel/Hankel-type envelope with the same large-$|\Delta t|$ decay scale, so that

$$C(\Delta t, 0) \sim (\text{oscillatory Bessel/Hankel prefactor}) \times e^{-\mu_\tau |\Delta t|} \quad \text{for} \quad |\Delta t| \gg \mu_\tau^{-1}, \tag{8.12}$$

up to representation-dependent prefactors and analytic continuation conventions. Appendix A derives the corresponding exponential envelope explicitly for spacelike separations; the present timelike statement is the detector-worldline analogue and is used here only at the level of its characteristic decay scale.

8.4 Closed-form TOA distribution: convolution formula

For a stationary Gaussian variable $X = \tau_\ell(t, x_D)$ with variance $\sigma^2_{\tau,\ell}$,

$$\langle \delta(t + X - \Theta) \rangle_X = \frac{1}{\sqrt{2\pi\sigma^2_{\tau,\ell}}} \exp\left[-\frac{(\Theta - t)^2}{2\sigma^2_{\tau,\ell}} \right]. \tag{8.13}$$

To obtain a closed convolution form, we further assume that, at leading order, the local clock fluctuation $\tau(t, x_D)$ is statistically independent of the matter-flux profile $\mathcal{F}(t)$. Physically, this corresponds to a weak-coupling or spectator-clock regime in which the clock sector broadens the readout without appreciably backreacting on the matter current. If clock–matter correlations are appreciable, mixed correlators contribute and the simple convolution formula below is replaced by a more general conditional-average expression.

Under the Gaussian and leading-order independence assumptions just stated, Eq. (8.9) becomes the convolution

$$p(\Theta) = \frac{1}{\mathcal{N}} \int_{-\infty}^{+\infty} \mathrm{d}t\, \mathcal{F}(t) \frac{1}{\sqrt{2\pi\sigma^2_{\tau,\ell}}} \exp\left[-\frac{(\Theta - t)^2}{2\sigma^2_{\tau,\ell}} \right]. \tag{8.14}$$

If $\mathcal{F}(t)$ is Gaussian as in (8.7), the convolution is again Gaussian:

$$p(\Theta) = \frac{1}{\sqrt{2\pi(\sigma_t^2 + \sigma^2_{\tau,\ell})}} \exp\left[-\frac{(\Theta - t_{\mathrm{cl}})^2}{2(\sigma_t^2 + \sigma^2_{\tau,\ell})} \right]. \tag{8.15}$$

Thus the clock sector produces a *predictive broadening*:

$$\mathrm{Var}(\Theta) = \sigma_t^2 + \sigma^2_{\tau,\ell}, \qquad \sigma^2_{\tau,\ell} = C_\ell(0,0). \tag{8.16}$$

Falsifiability.

Given an experimentally or numerically specified flux profile $\mathcal{F}(t)$ and a measured or modeled clock variance $\sigma^2_{\tau,\ell}$, the convolution formula (8.14) predicts a unique TOA distribution without additional fitting parameters. Deviations from the predicted variance $\mathrm{Var}(\Theta) = \sigma_t^2 + \sigma^2_{\tau,\ell}$ or from the expected crossover behavior at $\Delta t \sim \mu_\tau^{-1}$ would directly falsify the assumed clock sector or its effective field description.

Remark 8.1 (Gaussian tails vs exponential envelopes) The Gaussian form of $p(\Theta)$ arises from the *local, one-point* statistics of the clock readout at the detector. Exponential behavior enters instead through the *spacetime propagation* of clock correlations: early-arrival contributions requiring spacelike correlations are bounded by an exponential *envelope* set by μ_τ^{-1}, derived in Appendix A. These two effects are conceptually distinct and govern different regimes of the TOA distribution.

8.5 Discrete-event ultraviolet corrections

If the UV clock is an event-count $T_{\rm disc} = N_{\rm ev}$, then at time resolution $\Delta t \lesssim \ell$ one expects renewal-type corrections:

$$p(\Theta) \;=\; p_{\rm IR}(\Theta) \;+\; \delta p_{\rm UV}(\Theta), \tag{8.17}$$

where $p_{\rm IR}$ is the continuum prediction (e.g. (8.15)) and $\delta p_{\rm UV}$ carries integer-step features and non-Gaussianity that vanish upon coarse-graining $\Delta t \gg \ell$.

Bridge to a concrete realization. So far the discussion has been model-independent: J^μ and $j^\mu_{\rm ev}$ are treated as effective conserved structures. The next section simply records one explicit realization in which $j^\mu_{\rm ev}$ is generated by defect worldlines; none of the preceding definitions rely on that choice.

9 Example Microscopic Realization

The relational TOA construction above is model-independent: it requires only (i) a conserved matter flux J^μ through the detector world-tube Σ and (ii) a conserved (or approximately conserved) event current $j^\mu_{\rm ev}$ whose integral defines the ultraviolet event-count clock. Here we record one concrete realization in a hydrodynamic/defect setting motivated by Swirl-String Theory (SST), presented strictly as an example.

9.1 Defect (knot-center) worldline current

Consider N stable, localized defects with center worldlines $z^\mu_k(\tau)$. Define the distributional defect current

$$j^\mu_{\rm ev}(x) \;\equiv\; \sum_{k=1}^{N} q_k \int {\rm d}\tau\, \frac{{\rm d}z^\mu_k}{{\rm d}\tau}\, \delta^{(4)}\big(x - z_k(\tau)\big), \tag{9.1}$$

where $q_k \in \mathbb{Z}$ is an integer-valued defect charge. In the absence of creation/annihilation events in the bulk, this current is conserved in the distributional sense,

$$\nabla_\mu j^\mu_{\rm ev}(x) = 0, \tag{9.2}$$

and the associated event count through a detector world-tube Σ is

$$N_{\rm ev}(\Sigma) \equiv \int_\Sigma {\rm d}\Sigma_\mu\, j^\mu_{\rm ev}. \tag{9.3}$$

Operationally, $N_{\rm ev}(\Sigma)$ counts (with multiplicity q_k) the number of defect worldlines intersecting the detector region.

9.2 Continuum clock field from coarse-graining

At scales large compared to a coarse-graining length ℓ, define a smooth clock field $\chi(x)$ by

$$\partial_\mu \chi(x) \;\equiv\; \frac{1}{\nu_0}\left(j_{\mathrm{ev},\mu}(x)\right)_\ell, \tag{9.4}$$

where $(\cdot)_\ell$ denotes coarse-graining and ν_0 is a conversion factor. This yields the continuum-clock (IR) description of the event-count (UV) clock.

9.3 Foliation interpretation

If the macroscopic medium admits a hypersurface-orthogonal timelike congruence u_μ (a foliation), and if $\nabla_\mu \chi$ is everywhere timelike in the region of interest, then one may parametrize it by a scalar potential χ via

$$u_\mu \;=\; \frac{\nabla_\mu \chi}{\sqrt{g^{\alpha\beta}\nabla_\alpha \chi\, \nabla_\beta \chi}}. \tag{9.5}$$

In this interpretation, the TOA observable $p(\Theta)$ defined in Eq. (6.1) becomes the distribution of detector flux binned by the physical clock reading χ on Σ.

9.4 Interpretational note (optional)

In SST language one may view the localized defects as "topological matter excitations" in the effective medium. The integer charges q_k can then be interpreted as quantized defect labels. Classical hydrodynamic analogies (e.g., linking/helicity-based interpretations) are deferred to Discussion to avoid placing additional structure into the formal construction.

10 Discussion and Outlook (Analogy Only)

What the event current represents (and what it does not).

In the main text, the event current j^μ_{ev} is introduced as an (approximately) conserved quantity whose integral counts *localized, robust events* relevant to a detection process. At this level of generality it is deliberately agnostic: the "events" might be particle crossings of a detector region, interaction vertices effectively registered by an apparatus, defect/excitation worldlines intersecting Σ, or other countable processes whose total number is stable under smooth deformations of Σ that keep the boundary fixed.

This is the only structural assumption the formalism needs: that on the scales of interest the relevant events can be summarized by a current with $\nabla_\mu j^\mu_{\mathrm{ev}} \approx 0$ (exactly or with controlled violations). Section 9 provides one concrete realization in terms of defect worldlines in a hydrodynamic medium, but the relational TOA definition itself does not depend on adopting that ontology. Put differently, the construction is compatible with many microscopic stories; it only requires that they reduce to an effective event-count description on appropriate scales.

Classical analogy (optional intuition, not an assumption). The comparison with helicity in ideal fluid dynamics is meant as a guide to intuition rather than a

derivation. In fluids, conserved integrals can often be understood geometrically in terms of linking and knotting of flow structures [9–11]. By analogy, one may picture $j^\mu_{\rm ev}$ as a flux of "crossing events" that are topologically protected or otherwise robust, so that a discrete count becomes a smooth clock field under coarse-graining. None of the results above require this picture; it merely suggests one way in which a conserved event current might arise in an effective description.

Predictive content and where it enters. Once a clock sector is specified, its correlators $C(\Delta t, \Delta x)$ become physical input rather than interpretational decoration. In the simplest (Gaussian) setting they determine the amount of arrival-time broadening through local clock variance, while massive clock correlations also supply an exponential envelope for contributions that would require correlations across a spacelike separation (Appendix A). At sufficiently fine time resolution, the ultraviolet event-count description can additionally produce discrete/renewal features that are washed out in the infrared after coarse-graining.

Admissibility of event currents. The present formalism does not claim that every approximately conserved current is equally natural in a given experiment. The selection of an admissible $j^\mu_{\rm ev}$ is part of the effective model-building step and should be constrained by the detector context: locality, robustness under coarse-graining, compatibility with the detector's counting statistics, and empirical stability across changes of resolution. In this sense, the framework is not falsified or confirmed by a current in the abstract, but by whether a specified admissible current yields a clock sector whose correlators and TOA statistics agree with observation. A systematic classification of admissible event currents is left for future work.

Idealized measurement scenarios

A minimal scenario is a relativistic wavepacket experiment in which the detector couples (directly or effectively) to a clock degree of freedom with a finite correlation time. The observable consequence is that the TOA statistics need not coincide with those obtained from the matter wavepacket alone. In the Gaussian clock limit discussed in Sec. 8, the predicted broadening

$$\mathrm{Var}(\Theta) = \sigma_t^2 + \sigma_{\tau,\ell}^2 \tag{10.1}$$

would appear as an excess arrival-time spread beyond the semiclassical flux width σ_t.

From a practical standpoint, the framework is well-suited to numerical study. Given a flux profile $\mathcal{F}(t)$ and a model (or measurement-informed ansatz) for the clock correlator $C(\Delta t, \Delta x)$, the TOA density follows directly from the convolution formula, Eq. (8.14). This makes the clock-sector parameters (e.g. c_τ and μ_τ in the quadratic EFT) quantitatively testable in toy models and in lattice or continuum simulations.

11 Conclusion

We defined time-of-arrival (TOA) as a *relational*, manifestly covariant observable: detector crossings are binned neither by an externally supplied time coordinate nor by

postulating a universal time operator, but by a *physical clock* evaluated on the detector world-tube Σ, within the class of clocks that admit a (conserved or approximately conserved) event current. The construction uses two ingredients:

(i) a conserved matter current J^μ whose contraction with the unit normal defines the detector flux $\mathcal{F} = n_\mu J^\mu$, and
(ii) an event current $j^\mu_{\rm ev}$ whose integral defines a discrete event-count clock. The choice of $j^\mu_{\rm ev}$ is part of the effective model specification rather than an unobservable convention: different admissible event currents define different clock sectors and are, in principle, empirically distinguishable through the resulting TOA statistics.

A controlled coarse-graining over a scale ℓ then yields an infrared clock field $T(x)$ through

$$\partial_\mu T(x) = \frac{1}{\nu_0}\big(j_{{\rm ev},\mu}(x)\big)_\ell, \tag{11.1}$$

so that discrete counts and continuum time arise as UV and IR descriptions of the same underlying structure.

The resulting TOA density,

$$p(\Theta) = \frac{1}{\mathcal{N}} \int_\Sigma \mathrm{d}^3\sigma \, \sqrt{|h|}\, \mathcal{F}(\sigma)\, \delta\big(T_\Sigma(\sigma) - \Theta\big), \tag{11.2}$$

depends on detector microphysics only through the geometry of Σ, the effective flux J^μ, and the clock-sector correlators. Because the definition never requires a self-adjoint operator conjugate to the Hamiltonian, it sidesteps Pauli-type obstructions at the outset. In the semiclassical regime it reproduces the usual time-of-flight notion.

In the worked $1+1$D example, a narrowband incoming wavepacket together with a stationary Gaussian clock sector reduces $p(\Theta)$ to a convolution of the semiclassical flux profile with the local clock readout distribution, producing a concrete prediction: arrival-time broadening

$$\mathrm{Var}(\Theta) = \sigma_t^2 + \sigma_{\tau,\ell}^2, \qquad \sigma_{\tau,\ell}^2 = \langle \tau_\ell^2 \rangle, \tag{11.3}$$

with $\sigma_{\tau,\ell}^2$ fixed by the clock sector together with the detector/coarse-graining scale. Beyond this local effect, Appendix A shows that massive clock correlators are exponentially suppressed at spacelike separations with envelope scale μ_τ^{-1} (up to algebraic prefactors), providing a controlled bound on early-arrival contributions without implying superluminal signaling.

Relative to mainstream TOA formalisms, the point here is not that detector models or POVMs fail operationally, but that the relational definition isolates a covariant field-functional whose dependence on measurement details is pushed into explicit, modelable structures: conserved currents and a clock sector with its own dynamics. This reframes detector dependence as a UV→IR matching problem—event counts to continuum time—making TOA a covariant, predictive field observable.

Next steps

Immediate extensions include: (i) a $3+1$D worked example for finite-size detectors to quantify geometric dependence; (ii) non-Gaussian clock dynamics beyond the quadratic EFT to predict higher-cumulant signatures in $p(\Theta)$; and (iii) a systematic treatment of renewal-type UV corrections at temporal resolution $\Delta t \lesssim \ell$ and their crossover to the continuum limit. In particular, a numerical study of Eq. (8.14) as μ_τ is varied would directly probe the predicted exponential suppression envelope and the crossover from discrete event-count statistics to continuum clock time.

Appendix

A Exponential Suppression from the Clock Propagator

This appendix derives the exponential suppression scale governing spacelike clock correlations and motivates the corresponding exponential envelope expected in early-arrival tails for models in which such tails are controlled by clock-sector correlations. The derivation uses standard massive-scalar correlators and Bessel function asymptotics [6–8].

A.1 Clock correlators from the quadratic EFT

Consider the quadratic clock action in flat $1+1$D with $c_\tau = 1$:

$$S_\tau = \frac{1}{2}\int \mathrm{d}^2x \left[(\partial_t\tau)^2 - (\partial_x\tau)^2 - \mu_\tau^2\tau^2\right]. \tag{A.1}$$

The Wightman function $G^+(x)$ for a free massive scalar is a Lorentz-invariant function of $s^2 = (\Delta t)^2 - (\Delta x)^2$:

$$G^+(\Delta t, \Delta x) = \langle \tau(t,x)\tau(t+\Delta t, x+\Delta x)\rangle. \tag{A.2}$$

For spacelike separations $(\Delta x)^2 - (\Delta t)^2 > 0$, standard results give (see, e.g., representations in [7] and integral tables):

$$G^+(\Delta t, \Delta x) = \frac{1}{2\pi} K_0\Big(\mu_\tau \sqrt{(\Delta x)^2 - (\Delta t)^2}\Big) \quad \text{(spacelike)}. \tag{A.3}$$

While conventions vary by factors and $i0^+$ prescriptions, the key fact is that the spacelike decay is controlled by K_0.

A.2 Bessel asymptotics and exponential decay

For large argument $z \gg 1$, the modified Bessel function satisfies

$$K_0(z) \sim \sqrt{\frac{\pi}{2z}}\, e^{-z}\left[1 + \mathcal{O}\left(\frac{1}{z}\right)\right]. \tag{A.4}$$

Therefore, for spacelike separations with invariant distance

$$\rho \equiv \sqrt{(\Delta x)^2 - (\Delta t)^2} \gg \mu_\tau^{-1}, \tag{A.5}$$

Eq. (A.3) yields the exponential suppression

$$G^+(\Delta t, \Delta x) \;\propto\; \frac{e^{-\mu_\tau \rho}}{\sqrt{\mu_\tau \rho}} \left[1 + \mathcal{O}\big((\mu_\tau \rho)^{-1}\big)\right]. \tag{A.6}$$

Proposition A.1 (Spacelike correlator envelope) *For the quadratic clock EFT, the spacelike two-point function is exponentially suppressed with characteristic envelope scale μ_τ^{-1} as in* (A.6).

Remark A.1 This mechanism is directly analogous to standard QFT statements: massive fields do not transmit strong correlations outside the light cone; the residual correlations decay as $e^{-\mu\rho}$ (up to power-law prefactors). This is a correlator statement and does not imply superluminal signaling.

A.3 Heuristic implication for early-arrival tails

We now state a physically motivated inference rather than a theorem. In any TOA model in which contributions to sufficiently early detector readout require clock-sector correlations across an effectively spacelike interval, the spacelike suppression (A.6) suggests an exponential envelope of the form

$$p_{\text{early}}(\Delta) \;\lesssim\; A(\Delta)\, \exp\big(-\mu_\tau\, \rho(\Delta)\big), \tag{A.7}$$

where Δ parameterizes the degree of earliness and $\rho(\Delta)$ denotes the corresponding spacelike separation scale.

Here $A(\Delta)$ is left unspecified; in concrete realizations it must be derived from the detailed coupling between the clock sector and the TOA statistic. Accordingly, Eq. (A.7) should be read as a heuristic envelope statement induced by the clock correlator, not as a model-independent theorem for all TOA constructions.

A.4 Retarded Green's function and locality

For completeness, the retarded propagator solves

$$(\Box + \mu_\tau^2)\, G_{\text{ret}}(x) = \delta^{(2)}(x), \qquad G_{\text{ret}}(x) = 0 \text{ for } t < 0. \tag{A.8}$$

Local detector couplings to τ then inherit causal support from G_{ret}, while spacelike Wightman correlations remain exponentially suppressed as above. This distinguishes *correlation tails* from *signal propagation*, consistent with standard relativistic QFT [6, 7].

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Author contributions

O.I. conceived the study, developed the analysis, performed the derivations, and wrote the manuscript.

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